

# EVOARC

## PCB DESIGN CAPABILITIES

PCB Design | Hardware Development | Technical Consultancy

### Engineering Reliable. Designing Smarter.

EVOARC provides end-to-end PCB design services, from schematic and component selection through multilayer layout, high-speed routing, design verification, DFM/DFA and manufacturing release.

### PCB Design & Layout

- Schematic capture and design review
- 2-layer, 4-layer, 6-layer and multilayer PCB design
- High-density and mixed-signal layout
- Placement optimization and design-rule management

### High-Speed PCB Design

- Differential pairs
- Length matching and skew management
- Controlled impedance
- USB 2.0 / USB 3.x, Ethernet, MIPI, LVDS, PCIe\* and DDR\*

### Power Electronics

- DC-DC converters and LDOs
- High-current layout
- Power distribution
- Thermal management
- MOSFET / IGBT stages and protection

### Embedded & Controller Boards

- ARM / STM32 / ESP32
- Microprocessor systems
- FPGA\* / DSP\* / Embedded Linux\*
- Industrial controllers

### Communication Interfaces

- USB
- Ethernet
- CAN / CAN-FD
- RS-232 / RS-485
- UART / SPI / I<sup>2</sup>C / I<sup>2</sup>S
- GPIO / Modbus

### RF & Wireless

- Wi-Fi
- Bluetooth / BLE

- GNSS / GPS
- Cellular modules
- RF keep-out and grounding

## Sensor & Interface Boards

- Camera modules
- Image sensors
- IMUs
- Encoders
- Industrial sensors
- ADC / DAC

## Signal Integrity & EMI/EMC

- Stack-up and impedance planning
- Return paths
- Ground / power planes
- Crosstalk reduction
- Decoupling, filtering and noise reduction

## DFM / DFA

- Fabrication and assembly constraints
- Trace/space and drill rules
- Component spacing
- Fiducials
- Panelization
- Inspection and rework considerations

## Library & Component Management

- Symbols
- Footprints
- 3D models
- IPC-aware footprint development
- Datasheet validation
- BOM optimization

## PCB Review & Verification

- ERC
- DRC
- Connectivity
- Placement and routing review
- Power / ground review
- SI/PI, thermal and manufacturing review

# Advanced DFM / DFA Capability

## High-Layer-Count & Fine-Pitch BGA Design

EVOARC supports advanced PCB design and manufacturing-oriented layout for boards up to **42 layers** and fine-pitch BGA packages down to **0.4 mm pitch**, subject to the selected fabricator, stack-up, material system, via technology, component package and assembly process.

| Parameter             | EVOARC Capability                         | Key Considerations                               |
|-----------------------|-------------------------------------------|--------------------------------------------------|
| Layer Count           | Up to 42 layers*                          | Stack-up, sequential lamination and registration |
| BGA Pitch             | Down to 0.4 mm*                           | Escape routing, pad/via strategy and solder mask |
| BGA Escape            | Dog-bone / via-in-pad / microvia*         | Package geometry and routing channels            |
| HDI / Microvias       | HDI and microvia-aware layout*            | Laser drill, capture pad and sequential build-up |
| Trace / Space         | Fabricator-capability driven*             | Copper thickness, etching and impedance          |
| Via Technology        | Through / blind / buried / microvia*      | Aspect ratio, drill and annular ring             |
| Controlled Impedance  | Single-ended and differential             | Stack-up, dielectric, copper and geometry        |
| Copper / Planes       | Power and ground optimization             | Current, thermal and return paths                |
| Solder Mask           | Clearance and dam review                  | Fine-pitch BGA and dense SMT                     |
| Assembly / Inspection | SMT, THT, AOI and X-ray considerations    | Placement, solderability and hidden BGA joints   |
| Documentation         | Fab drawing, stack-up and release package | Manufacturing-ready communication                |

\*Final achievable dimensions and processes depend on the approved PCB fabricator/assembler.

## 42-Layer / 0.4 mm BGA Design Review Checklist

- Confirm approved 42-layer stack-up and material system.
- Define impedance-controlled layers, reference planes and critical routing constraints.
- Validate the 0.4 mm BGA footprint against the component datasheet.
- Select the BGA escape strategy and via technology with the fabricator.
- Check minimum trace/space, drill, annular ring, solder-mask and copper constraints.
- Review power delivery, plane transitions, stitching vias and return-current paths.
- Apply length matching and differential-pair constraints where required.
- Review thermal paths, copper spreading and thermal-via requirements.
- Complete DRC/ERC and final DFM/DFA review before release.

# EVOARC Engineering Toolset

Design • CAM • Simulation

## PCB Design Tools

| Tool            | Application / Capability                                               |
|-----------------|------------------------------------------------------------------------|
| Altium Designer | Schematic capture, PCB layout, multilayer and high-speed design        |
| Cadence Allegro | Advanced PCB layout, constraint-driven design and high-density routing |
| PADS            | Schematic and PCB design for professional electronics development      |
| KiCad           | Schematic, PCB layout and open-source design workflows                 |

## CAM / Manufacturing Tools

| Tool      | Application / Capability                                             |
|-----------|----------------------------------------------------------------------|
| CAM350    | CAM data verification, Gerber analysis and manufacturing data review |
| Camtastic | CAM inspection and PCB manufacturing data analysis                   |
| Valor NPI | DFM, manufacturing analysis and new-product introduction validation  |

## Simulation Tools

| Tool    | Application / Capability                                                                      |
|---------|-----------------------------------------------------------------------------------------------|
| ANSYS   | Engineering simulation and analysis for electrical, thermal and related design considerations |
| LTspice | Circuit simulation, power electronics analysis and analog/transient verification              |

Tool usage is selected according to project requirements, customer environment, manufacturing partner capability and the required design / verification workflow.

EVOARC • Engineering • Innovation • Execution